

Exercise 3 - Backpropagation

05.05.2026

This exercise must be uploaded as a handwritten solution (scan or photo). In total, the exercise gives 115 points. The percentages for each (sub)task are denoted in the title. Please leave us anonymous feedback for this exercise at <https://survey.mbp.tf.uni-bielefeld.de/index.php?r=survey/index&sid=576629&lang=en>.



Task 1: Gradients are Row Vectors (15%)

Consider a function $f : \mathbb{R}^{n \times 1} \rightarrow \mathbb{R}^{m \times 1}$, mapping n -dimensional column vectors to m -dimensional column vectors.

For any $\mathbf{x} \in \mathbb{R}^{n \times 1}$, the derivative $f'(\mathbf{x})$ is characterized by its linear approximation property:

$$f(\mathbf{x} + \mathbf{h}) \approx f(\mathbf{x}) + f'(\mathbf{x}) \cdot \mathbf{h}$$

for small $\mathbf{h} \in \mathbb{R}^{n \times 1}$.

Question:

Considering the dimensions of the involved entities $\mathbf{x}$, $\mathbf{h}$, $f(\mathbf{x})$, and $f'(\mathbf{x})$, explain why the gradient—i.e., the derivative of a scalar-valued function (when $m = 1$)—is a **row vector**.

Task 2: Gradient of a Bilinear Form (5% + 10% + 20%)

Let $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^m$ be column vectors.

The bilinear form is defined as

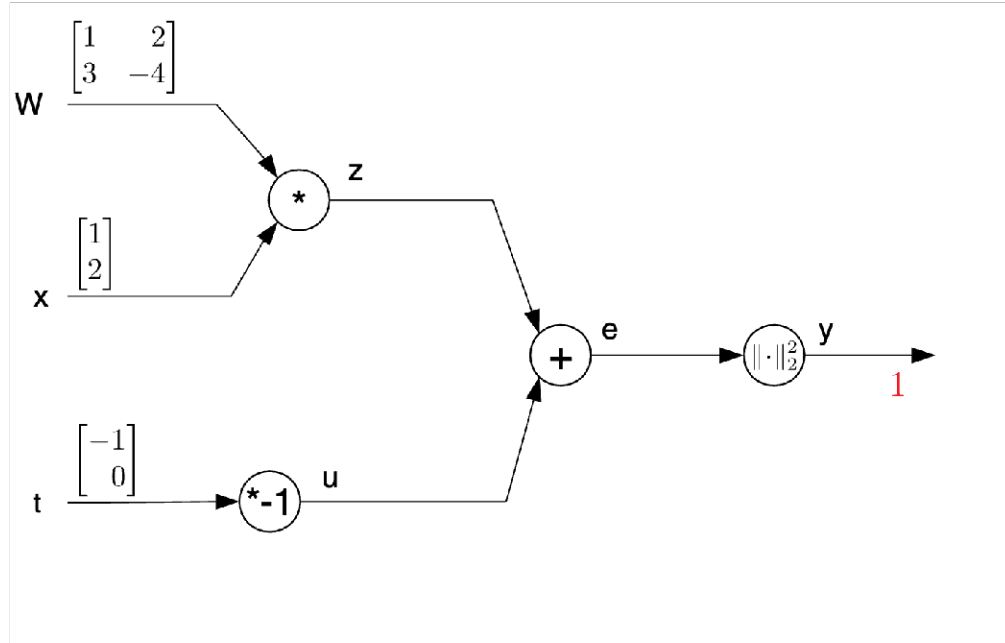
$$f(\mathbf{x}, \mathbf{y}, W) = \mathbf{x}^T W \mathbf{y} = \sum_i x_i \sum_j w_{ij} y_j$$

and yields a scalar output.

1. What is the correct dimension of the matrix W ?
2. Determine the dimensions of the following derivatives:
 - (a) $\nabla_{\mathbf{x}} f$ (standard row-vector gradient)
 - (b) $\nabla_{\mathbf{x}^T} f \equiv \nabla_{\mathbf{x}}^T f$ (column-vector gradient)
 - (c) $\nabla_{\mathbf{y}} f$ (row-vector gradient)
 - (d) $\nabla_{\mathbf{y}^T} f \equiv \nabla_{\mathbf{y}}^T f$ (column-vector gradient)
3. Compute the expressions for each of the derivatives listed above.

Task 3: Computation Graph (5% + 10% + 15% + 20%)

Consider the following computation graph:



1. Write the computation graph as a formula: $y =$
2. Perform a forward pass using the given values for $\mathbf{W}$, $\mathbf{x}$, and $\mathbf{t}$.
You can denote intermediate results directly in the graph above the connecting arrow lines if you printed this exercise.
3. Determine the local gradients of all operation nodes, in component-wise notation:

$$\begin{aligned}
 + : \quad & \frac{\partial e_i}{\partial z_j} = \delta_{ij}, \quad \frac{\partial e_i}{\partial u_j} = \delta_{ij} \\
 \|\cdot\|^2 : \\
 \times : \\
 \times -1 :
 \end{aligned}$$

4. Perform a full backward pass (backpropagation), computing the derivatives for all nodes. You can denote the results below the respective arrows (see red 1) if you printed this exercise.